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REMAGNETIZED CRATONIC CAMBRIAN STRATA FROM SOUTHERN NEVADA

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Abstract. Stratigraphic sections of Cambrian strata in southeastern Nevada are akin to thin, cratonic facies exposed in the Grand Canyon; their structural setting is much more complicated, however, from Mesozoic and Cenozoic tectonism. Paleomagnetic samples from two sections through these strata appear to have been completely remagnetized. Coarse-grained, light-colored sandstone from the Tapeats Sandstone yields scattered magnetizations, residing in hematite, that appear to reflect protracted diagenetic acquisition of remanence. A hematitic sandstone in the Bright Angel Shale yields relatively consistent 'Paleozoic' directions of magnetization, but petrographic study shows that the hematite results from diagenetic oxidation, and stratigraphic arguments suggest that the oxidation was not penecontemporaneous. Gray limestones of the Jangle and Muav Limestones yield a magnetization, residing in magnetite, that may reflect late Tertiary remagnetization, being imposed during uplift related to the onset of Basin and Range deformation. In any case, this magnetization differs greatly in direction from a hematite magnetization reported from slightly younger Muav Limestone in the Grand Canyon. Both the sampling sites also appear to have been tectonically rotated, but whether this rotation is true or a geometric artifact of the tilt correction cannot be determined from the present data. These results suggest that paleomagnetic data from rocks as old as Cambrian must be scrutinized very carefully before their magnetizations can be accepted as penecontemporaneous, and such scrutiny must include geologic data.

Introduction

Paleomagnetic work in Cambrian strata was undertaken in southern Nevada for several reasons. Elston and Bressler [1977] had presented paleomagnetic data from cratonic Cambrian rocks in the Grand Canyon on the Colorado Plateau, and Gillett and Van Alstine [1979] had begun paleomagnetic work in thick Cambrian miogeoclinal sequences farther to the west. In southeastern Nevada, stratigraphic sections of Paleozoic strata exist that are similar to the classic sequences of the Grand Canyon region exposed ~100 km to the east on the Colorado Plateau. It was thought that Cambrian sections in southern Nevada could provide a useful link for comparing paleomagnetic data from coeval strata in the geosyncline and on the craton. The Colorado Plateau region has nearly 'layer cake' stratigraphy; in contrast, the structure of southern Nevada is complicated from several episodes of Mesozoic and Cenozoic tectonism. Hence, it was also hoped that paleomagnetic data could help define the structural geology by constraining structural rotations, especially as Gillett and Van Alstine [1979] had inferred large amounts of vertical axis rotation resulting from related geologic structures farther to the west. Finally, although more paleomagnetic data from the Cambrian of North America have been reported in recent years, this part of the apparent polar wander (APW) path remains poorly known; pole positions are widely scattered with no clear age trends [e.g., Gillett and Van Alstine, 1979; Watts et al., 1980a, b]. As will be shown, none of these hopes was realized; however, the magnetic history that emerged serves to sound a useful cautionary note in paleomagnetic studies.

Geologic Setting

The region in which the paleomagnetic sections were sampled, in southern Nevada, is complicated by late Mesozoic and late Cenozoic tectonism (Figure 1). Mesozoic activity is represented by major, imbricate thrust sheets of the Late Cretaceous Sevier Orogeny that are exposed somewhat north and west of the sampling area. Cenozoic tectonism is related to the onset of Basin and Range extension and block faulting in the late Tertiary. In this area, Basin and Range structure is particularly complicated as it is also marked by major transcurrent faulting; southern Nevada lies in an east-west boundary area within two subprovinces of the Basin and Range [Anderson et al., 1972; Anderson, 1973]. On the northern side, Basin and Range activity is younger, and fault block mountains have large relief and thick Paleozoic sections. To the south, Basin and Range topography is more subdued, and thick sections of Tertiary volcanics rest directly on Precambrian crystalline rock. The details of the structure of this boundary area and the significance of the transcurrent faulting are still a subject of intense research. Current thinking has been well summarized by Bohannon [1979], and the description below is drawn from his presentation.

The Las Vegas Shear Zone, a zone of major late Tertiary right-lateral displacement, occurs on the western side of the region; it apparently offsets the Sevier thrusts about 100 km, between the Muddy Mountains on the north and the Spring Mountains on the south (Figure 1). Northeast trending, dominantly left-lateral strike slip faults of the Lake Mead Fault System [Bohannon, 1979] occur east of the shear zone. Finally, the

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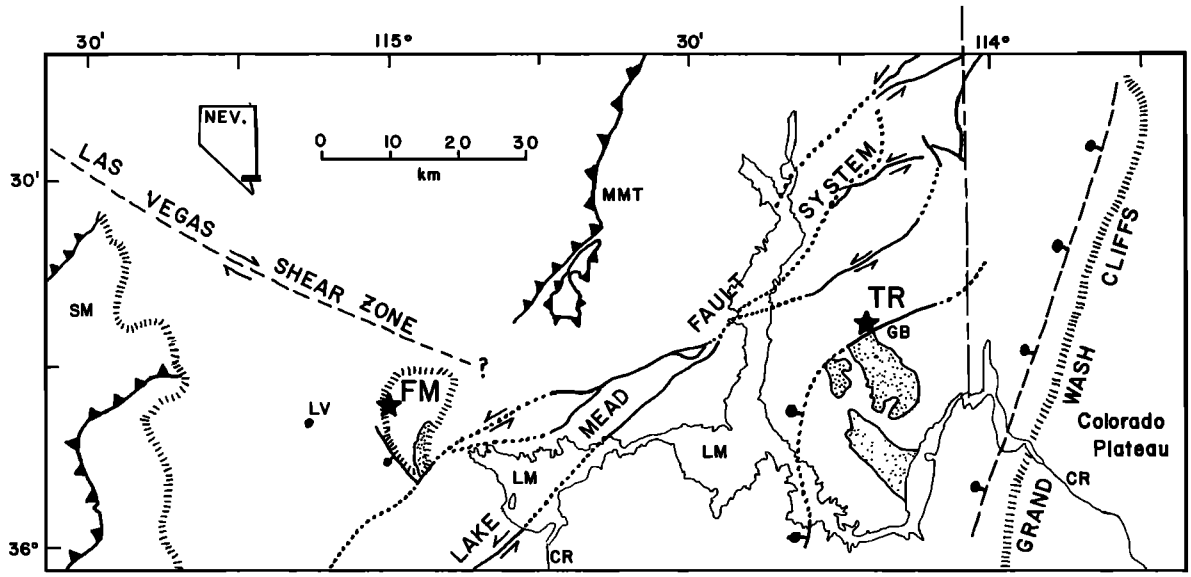


Fig. 1. Generalized tectonic map of part of southern Nevada and adjacent Arizona, showing locations of Cambrian sections that were sampled for paleomagnetic study (large stars). TR, Tramp Ridge; FM, Frenchman Mountain. Other symbols are as follows: LV, city of Las Vegas; CR, Colorado River; LM, Lake Mead; SM, Spring Mountains; MMT, Muddy Mountains Thrust; GB, Gold Butte fault. Heavy lines represent major faults: balls show downdropped side of normal faults; arrows indicate sense of displacement on strike slip faults; and teeth are on upper plate of thrust faults. The shaded areas south of Tramp Ridge indicate where the rapakivi-textured Gold Butte Granite crops out; similar shading on the east side of Frenchman Mountain indicates the area in which fanglomerates containing clasts of this granite occur. The vertical dashed line just west of longitude 114°W is the Arizona-Nevada state line. Adapted from Longwell et al. [1965] and Bohannon [1979].

Grand Wash Cliffs at the eastern edge of the region mark the boundary between the stable Colorado Plateau and Basin and Range normal faulting; the Grand Canyon ends where the Colorado River emerges from the Grand Wash Cliffs (Figure 1).

Two sections were sampled for paleomagnetic study, both in fault blocks of Paleozoic strata in which Cambrian rocks rest with profound unconformity on Precambrian crystalline basement. The westernmost section is at Frenchman Mountain (Figure 1), which is an east-dipping, homoclinal fault block [Longwell et al., 1965; C. R. Longwell, unpublished mapping]. Frenchman Mountain is out of place. On the east side of the mountain, large clasts of Gold Butte Granite, a Precambrian granite with distinctive rapakivi texture, are found in mid-Tertiary fanglomerate deposits. This granite crops out in place only at Gold Butte, about 60 km to the east (Figure 1); hence, Frenchman Mountain appears to be displaced westward by at least this amount. Additional evidence of major displacement arises from a striking contrast between the Paleozoic section at Frenchman Mountain and that in other ranges nearby. The strata at Frenchman Mountain represent a thin, typically cratonic section (like that near Gold Butte), whereas Paleozoic sections in the ranges immediately north and west are extremely thick and typical of the geosyncline. Details of the displacement are still controversial [Anderson, 1973; Longwell, 1974; Bohannon, 1979], but it is apparently related to the Tertiary transcurrent faulting. Thus, Frenchman Mountain may also have undergone substantial vertical axis rotations, and this must be remembered in interpreting the magnetic data.

The second section sampled for paleomagnetism is at Tramp Ridge, in east dipping strata just north of the Gold Butte fault (Figure 1). The latter fault is probably dip slip [Bohannon, 1979] so probably the section at Tramp Ridge has not undergone much overall vertical axis rotation; in detail, however, many minor faults cross the area and may have caused local tectonic rotations [Longwell et al., 1965, pl. 12].

As mentioned, Cambrian strata at Frenchman Mountain are very similar to the classic Grand Canyon section described by McKee and Resser [1945]. However, the stratigraphic nomenclature at Frenchman Mountain is rather confused for two reasons. First, the somewhat less appropriate terminology of the Pioche district in east-central Nevada has been applied to Frenchman Mountain [e.g., Longwell et al., 1965]. Second, facies changes complicate the relation of the strata to the Grand Canyon section. The stratigraphic relations will be set forth in some detail, as they are important for time constraints (Figure 2).

The basal unit at both Frenchman Mountain and Tramp Ridge is a thin, coarse-grained, quartzose sandstone that is correlated with the Tapeats Sandstone of the Grand Canyon [Longwell et al., 1965; Stewart, 1970]. It rests unconformably on crystalline rock. The Tapeats is overlain gradationally by a green, platy shale that correlates with the Bright Angel Shale of the Grand Canyon. At Frenchman Mountain, trilobites of late Early Cambrian age are found in the Bright Angel shale

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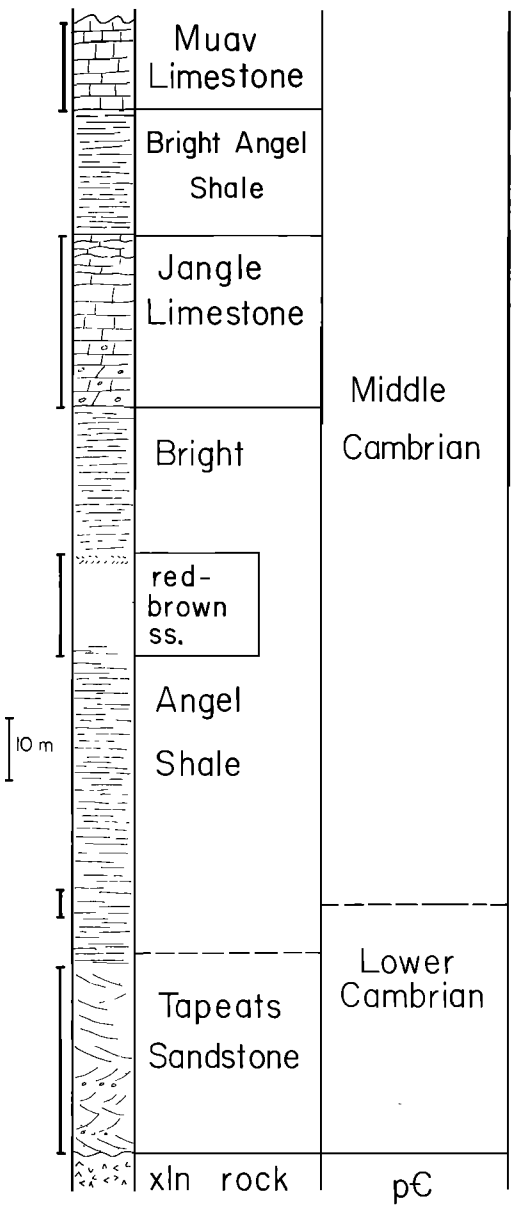


Fig. 2. Generalized stratigraphic column of the sampled sections of Cambrian strata in southern Nevada. Solid vertical bars alongside the stratigraphic section indicate units sampled for paleomagnetism.

immediately overlying the Tapeats [Pack and Gayle, 1971; A. R. Palmer, personal communication, 1978], which shows the Tapeats to be entirely of Early Cambrian age here; this also applies at Tramp Ridge.

A fine-grained, well-sorted, quartzose, reddish-brown sandstone occurs about 70 m above the base of the Bright Angel (Figure 2). On the basis of stratigraphic position and lithologic similarity, this unit is correlated with the 'red-brown unit' of McKee and Resser [1945] that occurs in the Bright Angel Shale in the western Grand Canyon. A. R. Palmer (personal communication, 1980) suggests that the red-brown unit may be the time equivalent of the Red Pass Limestone Member of the Carrara Formation; the Carrara crops out in the geosyncline to the west where Cambrian strata become much thicker. Palmer and Halley [1979] give a model for cyclic deposition in the Carrara in which thick limestone units represent relative marine stillstands, whereas intercalated shales and siltstones reflect relatively rapid rise in sea level. During a stillstand, while the carbonate bank developed offshore and prograded both landward and seaward, terrigenous sedimentation would prograde outward from the shore. Hence, a limestone unit intercalated in shale would be the time equivalent of a coarse terrigenous unit that was also sandwiched in shale. The red-brown unit occurs at roughly the correct stratigraphic position to be coeval with the Red Pass Limestone on this model [Palmer and Halley, 1979]. Correlation with the Red Pass suggests that the red-brown unit is of early Middle Cambrian age, and this is supported by the faunal data at Frenchman Mountain.

A gray limestone about 35 m thick occurs about 25 m above the red-brown unit at both Frenchman Mountain and Tramp Ridge (Figure 2). Traditionally this unit has been referred to the Lyndon Limestone of the Pioche District on the basis of its paleontologic age and stratigraphic position. However, this limestone is not continuous with the type Lyndon although the ages are nearly the same (A. R. Palmer, personal communication, 1979). Its correlatives in the Grand Canyon proper are the Meriwitica and Tincanebits Tongues, both composed of rusty brown dolomite, which represent a facies change [Palmer and Halley, 1979]. Westward into the geosyncline the unit correlates with the Jangle Limestone Member of the Carrara Formation, and that name will be used here. The Jangle is of Middle Cambrian age; trilobites of the Albertella zone are found just below it, and Glossopleura zone trilobites are found in the shales above.

Above the Jangle another green, platy shale occurs, which can either be considered a tongue of the Bright Angel or as the 'Chisholm' Shale if the Pioche terminology is used. Finally, a thick unit of gray limestone overlies this shale (Figure 2). It is roughly correlative with the Muav Limestone of the Grand Canyon [Palmer and Halley, 1979]. Samples were collected only in the stratigraphically lowest part of the Muav, which probably lies in the Middle Cambrian Glossopleura zone.

Sampling, Laboratory, and Statistical Procedures

Oriented samples 2.5 cm in diameter were collected with a hand-held, gasoline-powered drill similar to that described by Doell and Cox [1967]. One or two specimens 2.5 cm long were trimmed from a core; one was measured initially, the duplicate specimens being used for additional demagnetization experiments as needed. Trimmed limestone specimens were rinsed briefly with dilute HCl before measuring to minimize possible contamination from the drilling and sawing. In addition, a few tens of unoriented blocks were collected as lithologic samples to study textural relations.

Magnetic measurements were made on an ScT 2 axis superconducting rock magnetometer interfaced with a computer, allowing immediate, on-line data reduction. Both systems used have a background noise level of less than 1×10^{-10} A m^2 (1×10^{-7}

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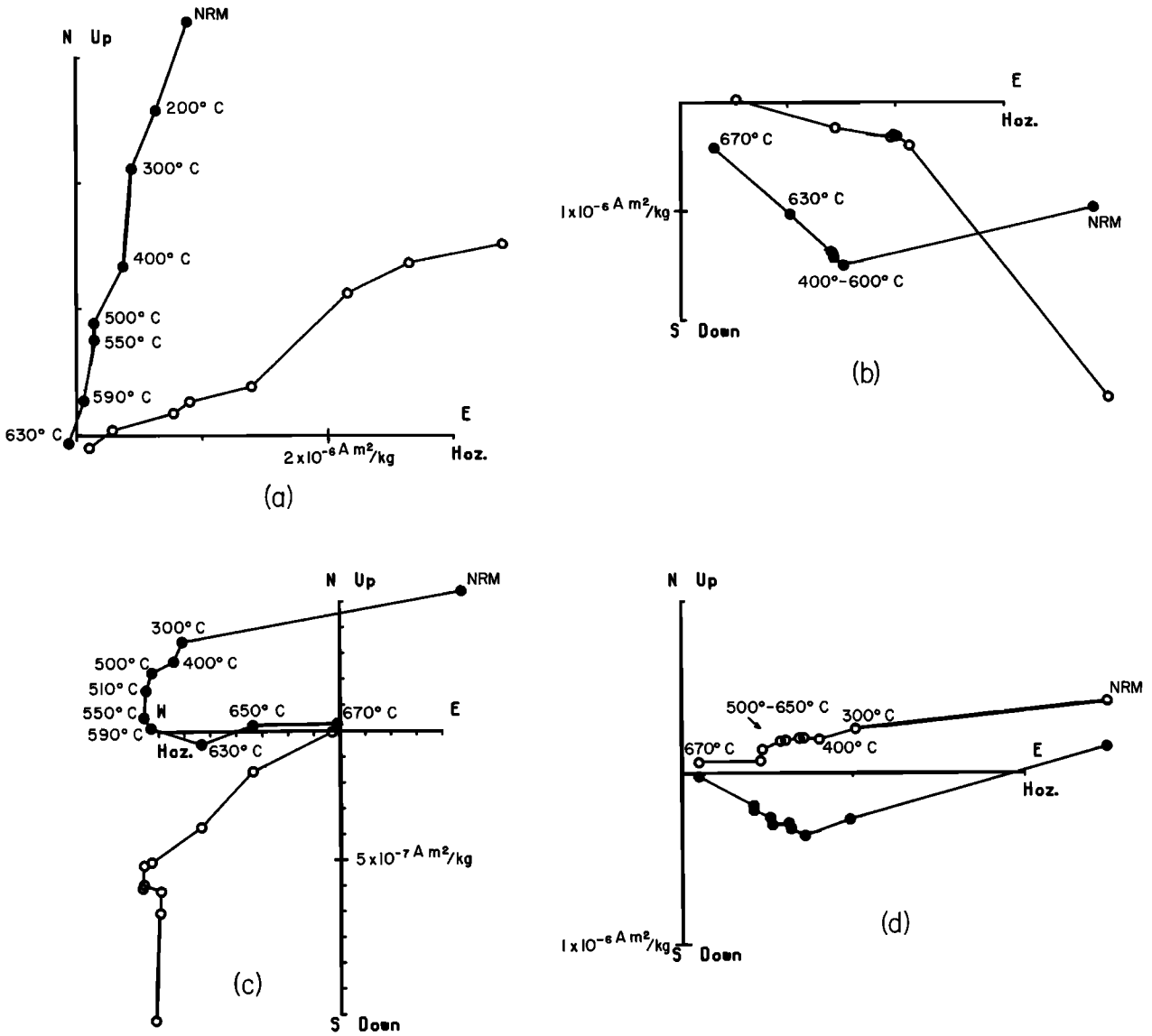


Figure 3. Vector demagnetization diagrams [Roy and Park, 1974] of samples from terrigenous strata, Cambrian of southern Nevada. Open (solid) circles represent inclination (declination). All directions are in stratigraphic ('tilt-corrected') coordinates. (a) Tapeats Sandstone, Frenchman Mountain. (b) Main part of red-brown unit of the Bright Angel Shale, Frenchman Mountain. (c) Red-brown unit, Tramp Ridge. (d) Red-brown unit, Tramp Ridge.

emu). Bulk susceptibility was measured on a number of limestone samples to monitor possible chemical changes occurring during thermal demagnetization [e.g., Turner, 1975]. Bulk susceptibility was measured using the cryogenic magnetometers [Gillett et al., 1982].

Progressive thermal and/or alternating field (af) demagnetization was performed on all samples. Thermal demagnetization is carried out with a μ-metal-shielded, noninductively wound electrical oven. When a run is completed, the samples are transferred to a cooling chamber, also shielded by μ-metal cylinders, in which the ambient field is <10 nT (1 nT = 1 γ). Specimens were always placed in nonsystematic orientations during demagnetization, so that systematic spurious components could not be imparted by stray fields. The af demagnetization was done with a Schoenstedt GSD-1 nontumbling demagnetizer.

The progressive demagnetization data from all specimens were plotted by computer, using a modification of Zijderveld's [1967] demagnetization diagrams similar to that employed by Roy and Park [1974]. Components of magnetization were identified visually as approximately linear stretches of points on the demagnetization diagrams or as clusters of consecutive points on the demagnetization path that showed little change in direction or intensity. Least squares fits were made to estimate these components [Gillett et al., 1982].

In many cases, modal analysis [Van Alstine, 1980] was employed on the distributions of directions, as, in general, they are skewed, probably because of the incomplete removal of secondary components. In these cases a mode is a better

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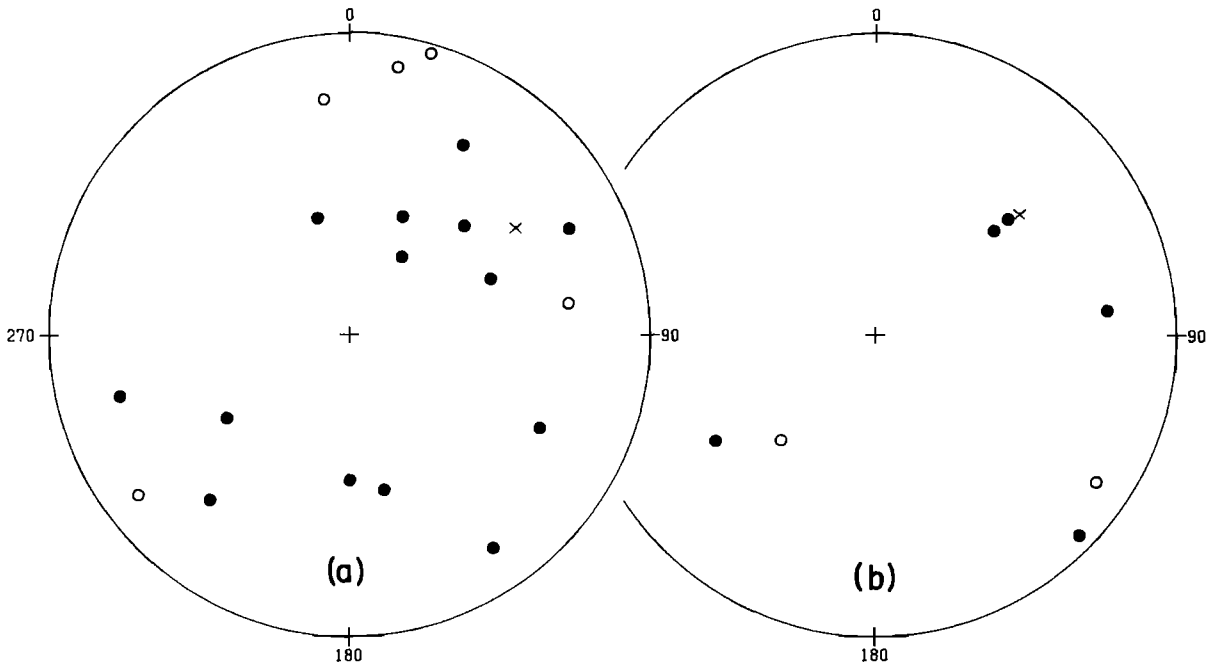


Fig. 4. Directions of characteristic magnetization, found by least squares fits to the demagnetization paths (see text), of samples from Frenchman Mountain. Equal-area projections in stratigraphic ('tilt-corrected') coordinates. Open (solid) symbols on upper (lower) hemisphere. Cross marks direction of present axial dipole field at sampling site after rotation for the mean attitude of the bedding. (a) Tapeats Sandstone. (b) Hematitic interbeds in lowest Bright Angel Shale.

estimate of central tendency than the mean. Bingham statistics [Bingham, 1974; Onstott, 1980] have been used as a guide to where modal analysis may be appropriate [Gillett et al., 1982], as they allow one to quantify whether a distribution departs significantly from circular symmetry about its mean.

To supplement the magnetic work, extensive petrography was carried out. At a minimum an unweathered, trimmed end from a core was polished and examined under a binocular microscope. For limestones, such a polished section was also etched briefly with dilute (~5%) HCl to highlight textural features. In addition, large polished sections were made from the blocks collected as lithologic samples. About 20 thin sections were also examined under reflected and transmitted light, and with cathodoluminescence, using a Nuclide Corporation Luminoscope. The emphases in the petrographic work were (1) to define the various lithologic types present, to see if there might be a correlation between rock type and magnetic behavior, and (2) to attempt to find textural features, including the mode of occurrence of magnetic minerals, that might help constrain the timing of the magnetization.

Results

Tapeats Sandstone

The Tapeats Sandstone was sampled only at Frenchman Mountain, where 19 samples were collected through the unit. Here, the Tapeats is a light-colored, coarse-grained, locally conglomeratic quartzose sandstone with abundant trough crossbeds. In polished sections, abundant evidence of pervasive, protracted oxidation is given by common spots several millimeters across that are composed of a diffuse mass of iron and probably manganese oxides. Grains that have been replaced with hematite or limonite are common and apparently reflect ferroan precursors.

Specimens from the Tapeats were thermally demagnetized at seven steps between 200° and 630°C. Most samples show a relatively steady decrease in magnetic intensity upon demagnetization with little change in direction (Figure 3a). Occasionally, a component aligned with the present field is removed below ~300°C. Evidently, the magnetic carrier is hematite as the magnetization persists to over 600°C. Although the direction from a given sample changes little, the overall distribution of directions is very scattered (Figure 4a).

Sandy Interbeds, Lower Bright Angel Shale

Seven samples were also taken in interbeds in the lower Bright Angel Shale at Frenchman Mountain. Four samples were collected in light-colored, quartzose interbeds, with oxidation spots, that greatly resemble the Tapeats Sandstone. Three samples were also taken in a distinctive, dark purplish-red sandstone to siltstone; the color arises from a pervasive, interstitial hematite pigment. Light red ooids, approximately 1 mm in diameter, occur in this unit. These ooids are now composed of hematite and clay minerals but probably were originally chamosite. Chamosite is an iron-bearing chlorite mineral [e.g., Hathaway, 1979], which is thought to be syngenetic and indicative of weakly oxidizing marine conditions [e.g., Porrenga, 1967] and

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which occasionally forms ooids [Pettijohn, 1957, pp. 454, 668]. Overall, the pervasive hematite cement appears to be a secondary feature superimposed on terrigenous marine rocks during some episode of oxidizing diagenesis.

All samples were thermally demagnetized at eight steps from 200° to 670°C. Their overall behavior, including the samples with the purplish pigment, is generally similar to the Tapeats Sandstone; the direction from a single sample shows little change with demagnetization, but the directions from all samples form a scattered distribution (Figure 4b).

Red-Brown Unit of the Bright Angel Shale

The red-brown unit of the Bright Angel Shale was sampled at both Frenchman Mountain and Tramp Ridge. At Frenchman Mountain, the sampled section is sandwiched between green shale; the main part of the unit is reddish, fine-grained sandstone, but the top part of the unit is light colored. A number of reddish sandstone interbeds also occur below the main sandstone ledge. Nine samples were collected in these interbeds, 19 in the main red sandstone body, and 4 in the upper, light-colored part. The unit is less well defined ~100 m laterally; more of the section is sandy, but the red color is more diffuse, shows more local variation, and there is much interbedded green shale.

At Tramp Ridge, the sampled section is again sandwiched between green shales, but it shows lateral and vertical color and grain size variations similar to those seen in the unsampled part at Frenchman Mountain. The section at Tramp Ridge is also thinner; 17 samples were collected through it.

Petrography of the red-brown unit. The main part of the section at Frenchman Mountain consists of fine-grained, well-sorted quartzose sandstone that commonly shows discontinuous planar lamination. Generally the rocks are brownish-red (5 YR 4/2) in color, but locally they have a purplish cast; the pigmentation arises from a microcrystalline, interstitial hematite cement. In thin section, this cement locally follows cracks within quartz grains, and common, dark, elongate aggregates of hematite parallel to bedding apparently reflect alteration of ferromagnesian precursors, perhaps biotite or amphiboles.

Glauconite peloids are present to common in this sandstone. They range in color from pale green to greenish black; many show partial hematitic alteration and some have been completely hematized, whereas others still have crisp unaltered edges. The glauconite is detrital in origin: the peloids show grain size sorting that reflects the size distribution of the surrounding quartz grains; occasionally the peloids are elongate parallel to bedding; variations in glauconite content commonly help define lamination within the sandstone. In thin section, interpenetrating relations between glauconite and quartz grains are present locally. Finally, the peloidal form is a common habit for glauconite grains [e.g., McRae, 1972].

A ferroan, rhombohedral carbonate (ankerite or ferroan dolomite) also occurs extensively in this red sandstone unit. The rhombohedra are orange and effervesce with cold, dilute HCl; apparently the crystals have been pervasively altered to goethite and calcite. Such calcite replacement ('dedolomitization') and concomitant iron oxidation is a common weathering phenomenon [e.g., Folkman, 1969; Braun and Friedman, 1970; Al-Hashimi and Hemingway, 1973]; similar weathering is ubiquitous in the Jangle and Muav Limestones, as will be described below.

At Frenchman Mountain, the reddish sandstone interbeds that occur below the main red sandstone unit greatly resemble it petrographically; they have a hematite cement, glauconite that commonly shows some oxidation, and oxidized rhombs of ankerite. The light-colored sandstone at the top of the Frenchman Mountain section, above the main red unit, lacks the hematite cement. However, these rocks also contain abundant glauconite and oxidized ankerite. Glauconite is more abundant in these light-colored beds; perhaps it has been better preserved because of the absence of subsequent hematitic alteration.

The sampled section at Tramp Ridge, as noted, has more variation of color than the one at Frenchman Mountain; the rocks range from brownish-red to buff or light greenish locally. Petrographic study shows many of the same features as seen at Frenchman Mountain; the color is due to a hematite cement, glauconite is common, and heavily oxidized rhombs of ankerite are found. Glauconite is more common in the light-colored parts of the section than in the red parts.

The red-brown unit at both Tramp Ridge and Frenchman Mountain has undergone an extensive diagenetic history characterized by widely varying redox conditions. As indicated above, the glauconite is evidently a primary detrital component. Glauconite is a syngenetic marine mineral that reflects weakly oxidizing conditions [Porrenga, 1967; Berner, 1971, pp. 200-201, 1981]; it is not in equilibrium with hematite. Therefore, the hematite must be secondary, as supported by the textural features described above: hematite occurs interstitially; 'ghosts' of replaced ferromagnesian grains occur; and more glauconite is present in the light-colored sandstones, suggesting that those rocks are less oxidized. (Although the glauconite is evidently the most resistant to hematization of the ferrous-iron-bearing minerals, it too has commonly been altered.) However, this hematite, although secondary, is probably not of recent origin. No hydrated or poorly crystallized ferric oxides, which would appear as orange rather than red pigment, apparently are present except those associated with the oxidation of the ferroan carbonate (discussed below). Hematite is the stable form of ferric oxide, but it does not nucleate easily due to large kinetic barriers. Unstable ferric oxyhydroxides, such as goethite, form first and these later age to hematite [e.g., Turner, 1980, pp. 338-344].

The rhombohedral ferroan carbonate (ankerite or ferroan dolomite) must have been formed after the hematization; had this carbonate been present when the hematite was formed, it would have undergone oxidation then. Carbonate minerals have relatively high solubilities (for geological substances) under all natural pH conditions. Ferrous iron that would be released by such

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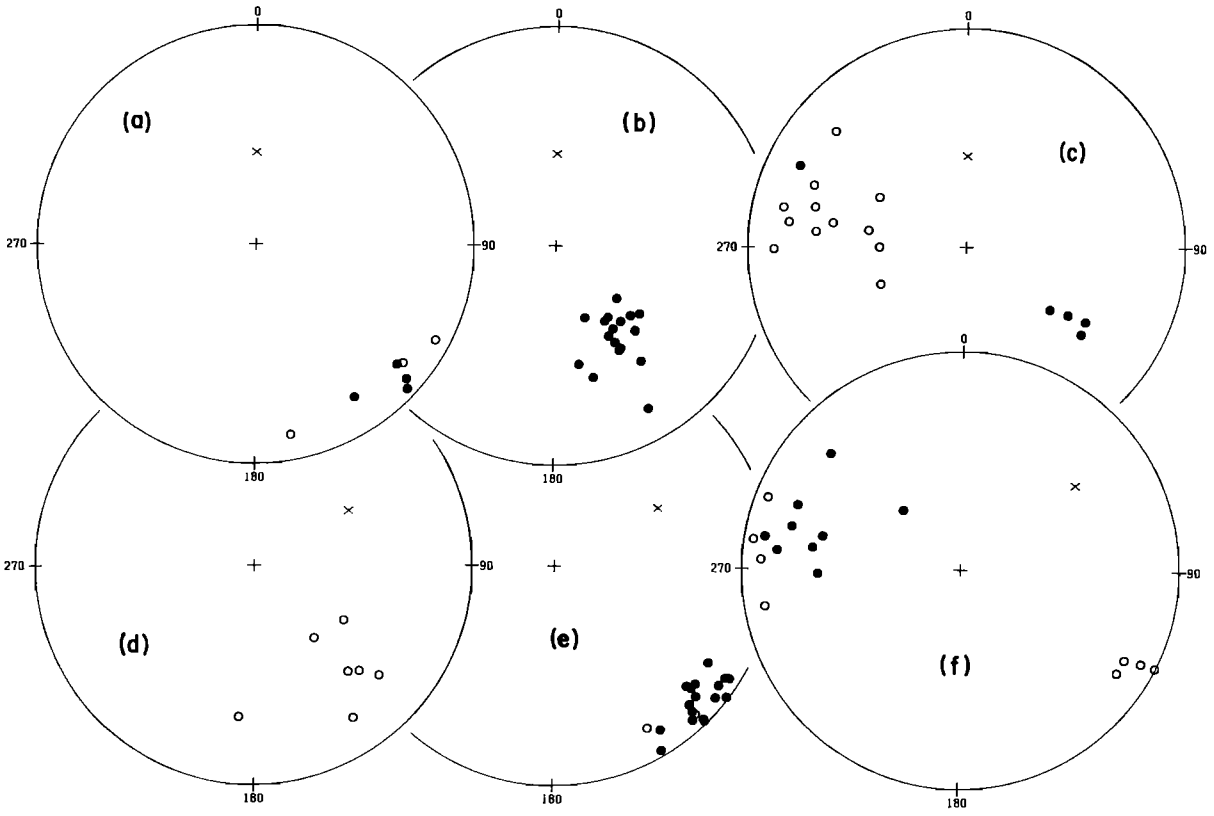


Fig. 5. Least squares fitted directions (see text) of characteristic magnetization for samples from red-brown unit of the Bright Angel Shale. (a)-(c) Geographic coordinates. (d)-(f) Stratigraphic coordinates. Other conventions as in Figure 4. (a) and (d) Basal red sandstone interbeds at Frenchman Mountain. (b) and (e) Main part of unit at Frenchman Mountain. (c) and (f) Red-brown unit at Tramp Ridge.

dissolution would undergo rapid oxidation if the groundwater were oxidizing. The oxidation of ferrous iron in solution happens very rapidly, occurring on a laboratory time scale. Thus, this carbonate must be an authigenic phase that reflects a return to more reducing conditions. An authigenic origin is also suggested by its euhedral, rhombic form. Further, the present oxidation of this carbonate probably reflects recent weathering, as the orange color of the oxidized rhombs suggests that the ferric iron mineral present is predominantly goethite, which is unstable with respect to hematite under virtually all geologic conditions [Berner, 1969, 1971, p. 197-199]; hence, early formed goethite would have inverted to hematite by now. It has been noted that ancient goethite is extremely rare [Fischer, 1963].

Paleomagnetism of the red-brown unit. All samples from the red-brown unit were subjected to progressive thermal demagnetization at at least six steps between 400° and 670°C. At Frenchman Mountain, the main, red sandstone part of the red-brown unit yields consistent behavior upon demagnetization. Thermal demagnetization to 400°C removes a component roughly reflecting the present field to reveal a characteristic magnetization with southeasterly declination and low, positive inclination. The direction and intensity change little upon further demagnetization to about 590°C, at which point the magnetization decays roughly toward the origin (Figure 3b). The characteristic magnetization has been estimated by least squares fits (Figures 5b and 5e) and a mode calculated (Table 1).

The samples from the basal, red sandstone interbeds at Frenchman Mountain also trend toward a southeasterly position with demagnetization; however, these directions have an inclination of ~-30° in stratigraphic coordinates (Figure 5d and Table 1). This direction differs significantly from that in the main red sandstone immediately overlying, despite the overall lithologic similarity of the two rock units.

Finally, the four samples from light-colored beds at the top of the Frenchman Mountain section yield only a present-field direction. These rocks lost the bulk of their NRM intensity upon the first thermal demagnetization step but were very resistant to af demagnetization, suggesting that their magnetization primarily resides in goethite from weathering.

The samples from Tramp Ridge yield more varied demagnetization behavior. These samples have a similar characteristic magnetization to that seen at Frenchman Mountain; however, two polarities are present. Most of the red-brown unit at Tramp Ridge has opposite polarity to Frenchman Mountain (Figure 3d); only the topmost few samples have similar polarity (Figures 5c and 5f). In addition, the Tramp Ridge samples have less uniform demagnetization behavior, although the light reddish and the red samples both show similar characteristic magnetizations. Many samples seem

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TABLE 1. Summary of Paleomagnetic Results From Cambrian Strata in Southern Nevada

Rock Unit (Abbreviation)	Average Attitude (Strike, Dip)	Geographic Coordinates D	I	k	α_{95}	N/N_{tot}	Stratigraphic Coordinates D	I	k	α_{95}	N/N_{tot}	North Pole Positions Direct Lat	Long	dp	dm	Rotated Lat	Long	Remarks
Frenchman Mountain (Lat. 36.19°N, Long. 115.01°W)																		
Red-brown unit, Bright Angel Shale																		
basal interbeds (FM_{r2})	22° 43°E	131°	4°	27	13°	6/7	136°	-36°	27	13°	6/7	48°N	145°E	9°	15°	17°N (39° ccw)	168°E	Mode not calculated because of small N
		(k_1 = -60, k_2 = -10.1)					(k_1 = -60, k_2 = -10.1)											
		(b^1 = 6.6*, g^2 = 27)					(b^1 = 6.7*, g^2 = 27)											
main part (FM_{r1})	19° 46°E	146°	47°	43	5°	19/19	133°	7°	45	5°	19/19	31°N	123°E	3°	5°	1°N (39° ccw)	150°E	
		(k_1 = -42, k_2 = -17)					(k_1 = -43, k_2 = -18)											
		(b^1 = 4.4, g^2 = 149)					(b^1 = 4.0, g^2 = 158)											
Jangle Limestone	4° 55°E	324°	51°	----	3°	32/47	40°	52°	----	2°	20/47	57°N	31°W	2°	3°			
		(k_1 = -27, k_2 = -12)					(k_1 = -26, k_2 = -11.4)											
		(b^1 = 8.7*, g^2 = 149)					(b^1 = 7.7*, g^2 = 214)											
Muav Limestone	5° 57°E	325°	46°	146	3°	15/15	38°	50°	133	3°	15/15	58°N	27°W	3°	4°			
		(k_1 = -100, k_2 = -66)					(k_1 = -80, k_2 = -66)											
		(b^1 = 0.7, g^2 = 485)					(b^1 = 0.2, g^2 = 483)											
Limestone Average (FM_ℓ)		---					---					58°N	29°W					
Tramp Ridge (Lat. 36.30°N, Long. 114.21°W)																		
Red-brown unit, Bright Angel Shale																		
normal	7° 50°E	283°	-31°	11	13°	13/13	282°	17°	11	13°	13/13	15°N	156°E	7°	13°			Mode not calculated because of uniform scatter
		(k_1 = -16.3, k_2 = -5.0)					(k_1 = -16.4, k_2 = -4.8)											
		(b^1 = 6.3*, g^2 = 24)					(b^1 = 6.9*, g^2 = 24)											
reversed		125°	41°	109	9°	4/4	119°	-10°	109	9°	4/4	26°N	142°E	5°	9°			Mode not calculated because of small N
		(k_1 = -652, k_2 = -39)					(k_1 = -633, k_2 = -39)											
		(b^1 = 14.9*, g^2 = 76)					(b^1 = 14.4*, g^2 = 76)											
Average, N & R (TR_r)		114°	-36°				111°	-14°				21°N	149°E			3°S	167°E	(30° ccw)
Jangle Limestone	8° 53°E	318°	45°	----	3°	10/17	34°	59°	----	3°	10/17	63°N	42°W	3°	4°			
		(k_1 = -86, k_2 = -20)					(k_1 = -86, k_2 = -20)											
		(b^1 = 11.3*, g^2 = 159)					(b^1 = 11.2*, g^2 = 159)											
Muav Limestone	8° 65°E	316°	33°	62	5°	17/18	31°	55°	62	5°	17/18	65°N	32°W	5°	7°			Mode calculated on basis of visual inspection; is preferred estimate
		(k_1 = -61, k_2 = -24)					(k_1 = -61, k_2 = -24)											
		(b^1 = 4.2, g^2 = 191)					(b^1 = 4.2, g^2 = 191)											
Mode:		313°	31°	----	3°	11/18	27°	58°	----	3°	14/18	69°N	39°W	3°	4°			
Limestone Average (TR_ℓ)		---					---					66°N	40°W					
Grand Canyon																		Grand Canyon data from Elston and Bressler [1977]
Tapeats Average (GC_t)		---					---					5°N	158°E	2°	3°			
Muav Average (GC_m)		---					---					55°N	110°E	2°	3°			

Abbreviation following unit name is label on Figure 9. Geographic (stratigraphic) coordinates are not (are) corrected for bedding dip. D, declination; I, inclination. The k (not given where modes calculated) is Fisher's [1953] concentration parameter. For α_{95}, where Fisher [1953] statistics calculated, half angle of 95% confidence; where modes calculated, estimate of half angle ('β_{95}') suggested by Van Alstine [1980]. N_{tot} is the total number of samples; for Fisher statistics, N is the number of samples in the mean; for modal analysis, N is the number of samples with a Fisherian mean equal to the mode; this mean is used to calculate β_{95} [Van Alstine, 1980]. 'Direct' pole position is calculated directly from stratigraphic coordinates; 'rotated' pole position is calculated for the red-brown unit after applying a counterclockwise vertical axis rotation (in parentheses) inferred from the direction of magnetization in the Jangle and Muav Limestones (see text). The dp (dm) is the semiminor (semimajor) axis of error ellipse. The k_1, k_2 are Bingham [1974] concentration parameters in Onstott's [1980] notation. The b,g are test statistics for circular symmetry [Mardia, 1972, pp. 277-278]; * indicates a significant departure from a circularly symmetric axial distribution at the 99% confidence level. Modes [Van Alstine, 1980] were calculated in these cases [Gillett et al., 1982].

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to have an additional, high-inclination component of magnetization that persists to high demagnetization steps (Figure 3c); this phenomenon may arise from superposition of both polarities over the same blocking temperature interval.

The best estimate of the mean direction from the red-brown unit at Tramp Ridge is probably an average of the mean of one polarity and the antipode of the mean of the other polarity (Table 1). As the normal and reversed directions are not antiparallel, this procedure tends to cancel any consistent bias from unremoved secondary components [e.g., McElhinny, 1973, p. 85].

The magnetic carrier in all the red units appears to be hematite, as in all cases the characteristic magnetization persists to over 600°C. Probably, this hematite is the pervasive pigment found in these rocks, as the pigment-free beds at Frenchman Mountain show no characteristic magnetization at all. Hence it follows that the characteristic magnetization was acquired subsequent to deposition, as it has been shown that this hematite is authigenic. Evidently, the acquisition of magnetization spanned at least one polarity reversal; the scatter observed between Frenchman Mountain and Tramp Ridge, and within the Frenchman Mountain section itself also suggests relatively protracted acquisition of remanence.

Jangle Limestone

Petrography of the Jangle. Forty-seven samples were collected through the Jangle Limestone at Frenchman Mountain. The lowest part of the unit here is mostly oolite, whereas the middle part is largely peloidal, ranging from mudstones to grainstones. The upper half of the Jangle appears to be mostly cryptalgal laminite [Aitken, 1967]; a wavy, laminated fabric is pervasive and commonly highlighted by dolomicrite. Birdseye fabric is common, and many of the birdseye voids appear to reflect desiccation as they are elongate and the upper and lower outlines of the voids roughly match, suggesting that they were pulled apart. Such rock types reflect common exposure to subaerial conditions and would be about the best candidates amongst limestones for penecontemporaneous oxidation and hematite formation.

Replacement of limestone by dolomite and ferroan dolomite is pervasive in all rock types. Several generations are present, some of which are manifested by rhombs up to 1 mm across. These very large rhombs are generally zoned by iron content; very commonly the ferroan zones have been oxidized to a mixture of goethite (or hematite) and calcite. Such alteration is very similar to that of the rhombohedral carbonate in the red-brown unit. Where unaltered, the ferroan dolomite has a slightly bluish cast; iron content was checked by staining with slightly acidified potassium ferricyanide [e.g., Wolf et al., 1967].

At Tramp Ridge, the Jangle is thinner, and only 17 samples were collected, mostly in lime mudstone. The section here is dominated by lime mudstone in the lower part and oolite in the upper part; no algal laminites were seen. Occasional dark red, hematitic blebs probably reflect pseudomorphs after pyrite on the basis of their form and observations in similar limestones elsewhere. Lime mudstones such as these reflect quiet water deposition; they are rich in organic matter, and reducing conditions prevail, so that early pyrite authigenesis (probably mediated by sulfate-reducing bacteria [e.g., Berner, 1981]) predominates. Unlike the algal laminites at Frenchman Mountain, early oxidation is not to be expected in such rocks. Diagenetic replacement by ferroan dolomite is also common in the Tramp Ridge section; ooids are commonly replaced by coarse aggregates of euhedral ferroan dolomite which has subsequently been oxidized and dedolomitized by weathering.

Paleomagnetism of the Jangle. The samples from both Frenchman Mountain and Tramp Ridge were subjected to thermal demagnetization at five steps from 200° to 450°C, and the demagnetization behavior of both sets of samples is very similar. The NRM directions reflect the present field at the site; after demagnetization at 200°C, only 25-35% of the NRM intensity remains and the sample directions move somewhat northwesterly, with steep positive inclination (Figure 6a). This direction shows little change with further demagnetization, the rocks becoming unmeasurable by 450°C. No correlation of magnetic behavior with lithologic type was found, despite the heterogeneity of the limestones represented.

Thermal demagnetization apparently removes a component that reflects the present field by 200°-300°C; removal at low temperatures suggests that, in part, this component resides in goethite from present weathering. The northwesterly magnetization revealed by demagnetization is not far from the present field, but it is distinct and apparently represents the characteristic magnetization of these rocks (Figure 7). The mean directions from Tramp Ridge and Frenchman Mountain are similar (Table 1).

To interpret this magnetization, it is important to determine the carrier of the magnetization. Combinations of demagnetization techniques allow one to constrain the magnetic carrier of a component of magnetization because a set of magnetic properties can be associated with that component, as has been noted by a number of authors [e.g., Roy et al., 1979; Roy and Anderson, 1981; Gillett et al., 1982]. Accordingly, five duplicate specimens were subjected to thermal demagnetization at 300°C, to remove the present-field component and then subjected to af demagnetization at five steps between 5 and 50 mT. The characteristic component decays upon af demagnetization (Figures 6b and 6c); therefore, the carrier in which it resides has a moderate coercivity and is probably magnetite. A very small amount of detrital magnetite could account for the intensity of magnetization, and these limestones are associated with terrigenous rocks so that a moderate detrital content is to be expected. Further, authigenic magnetite is not to be expected in the sulfur-rich environments of the lime mudstones at least [e.g., Berner, 1971, pp. 196-197, 1981].

Little change in susceptibility as a function of demagnetization step was found, but generally, it decreased somewhat. Hence, the magnetite is not an artifact of the heating, e.g., arising from in situ reduction [Shive and Diehl, 1977a, b]. Susceptibility would be expected to increase if magnetite had been formed during thermal demagnetization.

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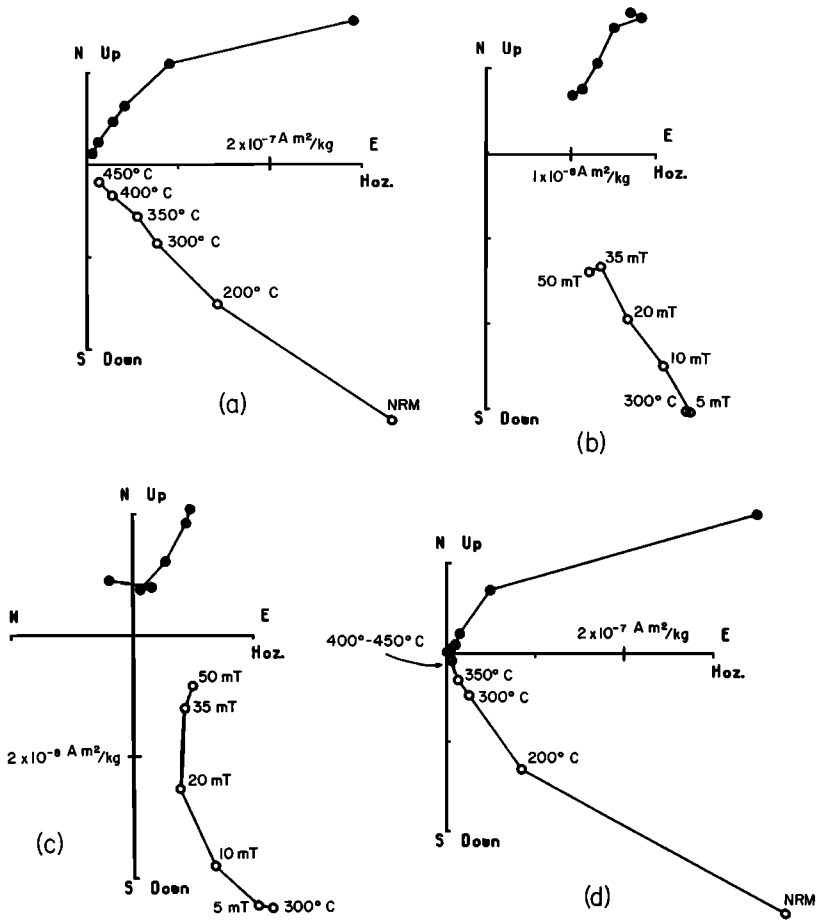


Fig. 6. Demagnetization behavior of Cambrian limestone samples from southern Nevada. (a) Thermal demagnetization, Jangle Limestone, Frenchman Mountain. (b) af-following-thermal demagnetization, Jangle Limestone, Tramp Ridge; note that NRM omitted. (c) af-following-thermal demagnetization, Jangle Limestone, Tramp Ridge; NRM also omitted. (d) Thermal demagnetization, Muav Limestone, Tramp Ridge. Conventions of diagrams as in Figure 3.

Finally, the demagnetization diagrams for the af experiments suggest that at least two additional components of magnetization are occasionally present in these limestones because the demagnetization paths do not trend toward the origin. In a few cases (Figure 6b) the magnetization stops decaying between 35 and 50 mT, suggesting that a weak, high-coercivity component is present. This component has high inclination, probably reflects the present field, and probably resides in hematite. In other samples (Figure 6c), the path appears to be trending toward a low-inclination position; such a magnetization may represent a Paleozoic, perhaps primary component. Low-inclination magnetizations have been found in similar limestones elsewhere [e.g., McElhinny and Opdyke, 1973; Gillett and Van Alstine, 1979; Gillett et al., 1982]. No direction could be calculated for this component in these strata, however, as only a vestige of magnetization is left.

Muav Limestone

Petrography of the Muav. At Frenchman Mountain, 15 samples were collected in the lowest 12 m of the Muav Limestone, whereas at Tramp Ridge 17 samples were taken in the lowest 18 m. In addition, at Tramp Ridge, three samples were collected in a fault breccia that is cemented with hematite-rich calcite.

The strata at both sections are very similar and resemble the Jangle Limestone at Tramp Ridge; the dominant rock type is dark brownish-gray lime mudstone that is locally peloidal. Hematitic blebs probably are pseudomorphs after pyrite. Much patchy recrystallization to microspar has apparently taken place. Burrows, reflected by spar-filled tubes, and stylolites are also common. Finally, replacement by ferroan dolomite is present to pervasive; the ferroan parts are commonly oxidized, but oxidation is less prevalent than in the Jangle Limestone.

Paleomagnetism of the Muav. The Muav samples from both Tramp Ridge and Frenchman Mountain were subjected to progressive demagnetization at five steps from 200° to 450°C; the three samples from the fault breccia at Tramp Ridge were additionally demagnetized up to 600°C. The breccia samples yield a magnetization with a direction very near that of the present field; as this magnetization persists to 600°C, it evidently resides in hematite. Probably, therefore, the fault is of geologically recent origin. Late

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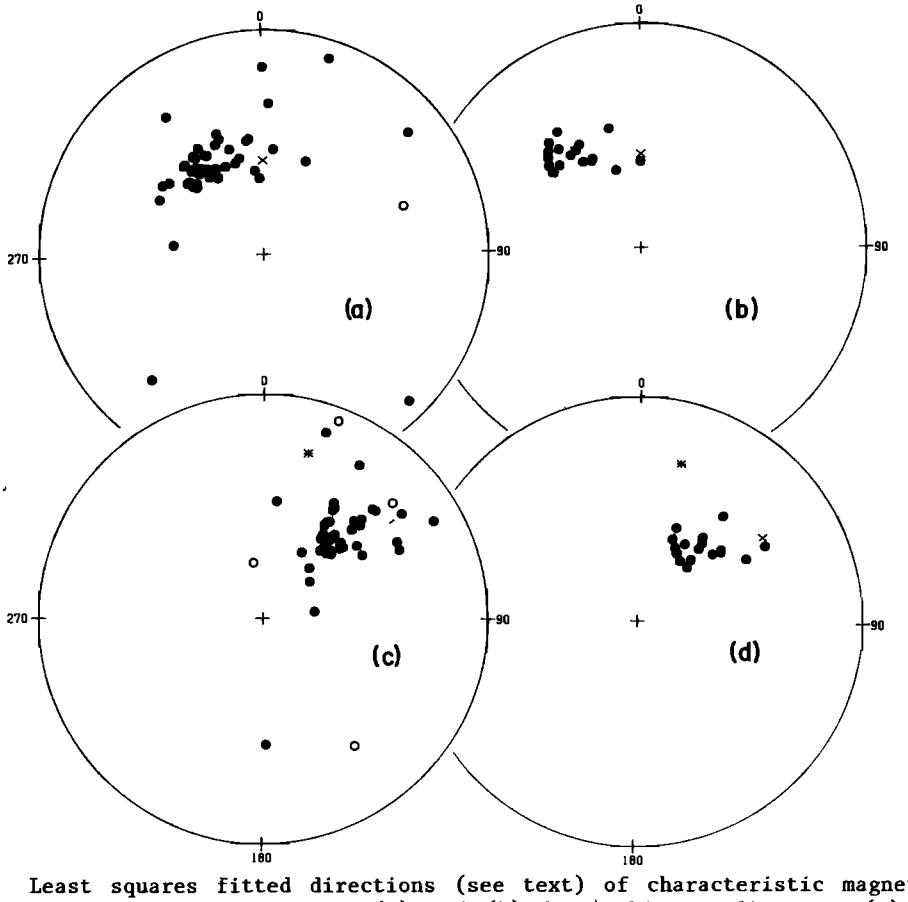


Fig. 7. Least squares fitted directions (see text) of characteristic magnetization from samples of Jangle Limestone. (a) and (b) Geographic coordinates. (c) and (d) Stratigraphic coordinates. Other conventions as in Figure 4. (a) and (c) Frenchman Mountain. (b) and (d) Tramp Ridge.

Tertiary faults that locally control mineralization are common in the Tramp Ridge area [Longwell et al., 1965, p. 126-128].

Results from the samples from the rest of the sections are very similar to those from the Jangle Limestone (Figure 6d). The NRM reflects the direction of the present field; after demagnetization at 200°C, the intensity drops to only ~30% of the NRM intensity, and the directions shift to a northwesterly position. Little change in direction occurs upon further demagnetization, and no consistent change in susceptibility with thermal demagnetization is found.

The mean direction from the Muav at the Frenchman Mountain section is very similar to those from both of the Jangle sections (Figure 8 and Table 1). However, both the mean and modal directions from the Muav at Tramp Ridge are slightly counterclockwise of those from the Jangle. This discrepancy appears to result from local, small-scale rotations of blocks where the Muav was sampled. The Muav was sampled along a ravine that cuts transversely across the strata, and the position of this watercourse is controlled by some minor faults. The measured dip of the Muav section is significantly steeper than the Jangle (Table 1), and these steeper dips probably result from minor rotations due to this faulting. The regional attitude is typified by that of the Jangle.

The nature of the magnetic carrier in the Muav was investigated using combinations of af and thermal demagnetization. Seven duplicate specimens were demagnetized at 300°C, and then af demagnetized at five steps to 50 mT. The magnetic behavior is very similar to that of the Jangle; apparently, the characteristic magnetization resides in magnetite. However, the demagnetization paths commonly do not trend toward the origin, suggesting that in some cases, vestiges of other components are present.

Discussion

To recapitulate, scattered directions of magnetization have been found in the Tapeats Sandstone and lower Bright Angel Shale; a low-inclination, two-polarity magnetization residing in hematite has been found in the red-brown unit of the Bright Angel Shale; and magnetite-borne magnetizations with steep positive inclination and northeasterly declination occur in both the Jangle and Muav Limestones. None of these characteristic magnetizations is a convincing candidate for a primary magnetization; their possible significance will be discussed below.

Limestone Units

The magnetization of the limestone units, although it apparently resides in magnetite, is probably not primary. The relatively high paleo-

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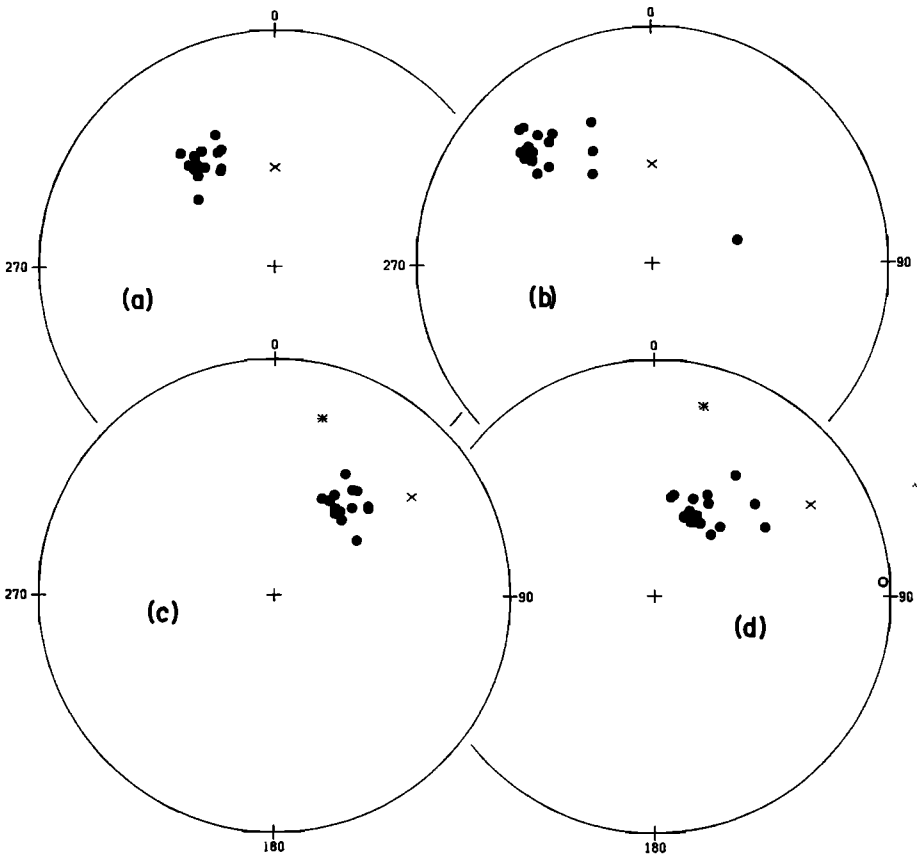


Fig. 8. Least squares fitted directions (see text) of characteristic magnetization from samples of Muav Limestone. (a) and (b) Geographic coordinates. (c) and (d) Stratigraphic coordinates. Other conventions as in Figure 4. (a) and (c) Frenchman Mountain. (b) and (d) Tramp Ridge.

latitude position (~38°) implied by the stratigraphic inclination is most likely too high for limestone deposition; shallow water marine limestones are formed in 'warm, shallow, clear water' [e.g., Wilson, 1975] in very low latitudes. Although it is possible that limestone deposition was less restricted to low latitudes during the Cambrian, because of less steep thermal gradients from the equator to the poles, this cannot be assumed a priori, and indeed evidence suggests that polar climates were cool during parts of the Cambrian at least. (It must be remembered, of course, that the Cambrian is about equal in duration to the entire Cenozoic, and blanket generalizations about 'Cambrian' climate are risky.) In the Early Cambrian (near the age of these strata), a relatively steep climatic zonation is suggested because the Early Cambrian was a time of relatively low sea level stand (rather like the Holocene) and the ameliorating effects of shallow seas were not present. Further, the distribution of Cambrian agnostid trilobites, a probable cold, deep-water fauna, is probably similar to modern isopod crustacea [Taylor and Forester, 1979]; in addition, abrupt extinctions between trilobite faunal units ('biomeres' of Palmer [1979]) throughout the Cambrian may be related to invasion of cold water onto the continental shelf [Stitt, 1977]. In both these cases, cold deep marine water implies cool poles. Finally, preliminary paleoclimatic analysis of the Late Cambrian [Ziegler et al., 1981] suggests that climatic zonation then was even more extreme than today. In sum, it seem likely that climatic control on limestone deposition in the Cambrian was not greatly different from that today. In addition, the relatively low coercivities that are implied by the af demagnetization behavior as well as the vestige of a low-inclination component in some specimens also suggest that the magnetization is secondary.

The dip-corrected directions from both Frenchman Mountain and Tramp Ridge have northeasterly declinations, however, which does not correspond to any point on the post-Cambrian North American polar wander path, although their inclinations are roughly consistent either with a Cretaceous or Tertiary geomagnetic field. Therefore, it seems possible that the directions reflect tectonically rotated Cretaceous or Tertiary positions. As both the Cretaceous and Tertiary were times of major tectonic activity in the general area, such rotation and remagnetization is also reasonable on geologic grounds. The most likely mechanism for imposing such remagnetization would be heating, causing a partial thermoremanent magnetization (PTRM) to be acquired when the strata cooled later. Times of regional tectonism, involving higher heat flow and subsequent uplift, are thus favorable periods for remagnetization.

The Tertiary seems to be a more likely time for remagnetization. These strata are in the foreland east of the Cretaceous Sevier thrusting, and it seems less likely that they would have been

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affected by the thrusting. In contrast, Tertiary Basin and Range tectonism has had profound effects in the area, as outlined in the section on the geologic setting. Higher-than-average heat flow is also characteristic of the Basin and Range province [e.g., Eaton, 1980] and regional uplift has accompanied Basin and Range block faulting [e.g., Eaton et al., 1978; Eaton, 1979; Lucchita, 1979]. Finally, the magnetization direction of the Muav and Jangle Limestones is nearer a Tertiary position, so that the postulated tectonic rotations are minimized.

If these directions reflect Tertiary remagnetization, both sections show apparent clockwise rotations; the Tramp Ridge strata are about 30° from due north, whereas the Frenchman Mountain sections show a 39° difference. It is curious that Tramp Ridge shows nearly as large a declination discrepancy as does Frenchman Mountain. As mentioned in the geologic-setting discussion, large, consistent vertical-axis rotations would not be expected at Tramp Ridge as this section is far from the transcurrent faults. Perhaps the tectonic rotation is a local effect resulting from the many minor faults present.

As emphasized by MacDonald [1980], however, declination anomalies may not always reflect true rotation about a vertical axis but may be geometric artifacts of the tilt correction. This mechanism potentially applies both at Frenchman Mountain and Tramp Ridge, and distinguishing true from apparent vertical axis rotations is not possible with the present data. These discrepancies, however, underscore the ambiguity that can arise with paleomagnetic data from structurally complicated areas.

Elston and Bressler [1977] reported paleomagnetic data from slightly younger Muav strata in the Grand Canyon and found very different results from those obtained here. The rock types they sampled there are 'hematite laminated carbonate' and 'red sandy carbonate,' the latter thought to reflect deposition during a marine regression. The characteristic magnetization in those strata has southeasterly declination, shallow negative inclination, and resides in hematite (Table 1 and Figure 9); it is based on bulk demagnetization of all samples at 500°C after progressive thermal demagnetization of pilot specimens. This direction is nearly identical to that of a characteristic magnetization, also residing in hematite, from the Devonian Temple Butte Limestone that unconformably overlies the Muav in the Grand Canyon. In turn, both magnetizations are very similar to a 'late Paleozoic' magnetization.

These authors believe the time of oxidation and hematite formation in both formations to be early and cite some field evidence but do not report any petrographic observations. Supratidal facies, in which early oxidation seems more likely, have been described from parts of the Muav [Wanless, 1975]. However, whether such a facies was sampled by Elston and Bressler is not certain, and in any case, such supratidal facies in the Jangle Limestone at Frenchman Mountain yield results no different from the other limestone types, as has been described.

Possibly oxidizing diagenesis later in the Paleozoic has destroyed all reduced minerals in the Muav in the Grand Canyon. Petrographic investigation of the mode of hematite occurrence

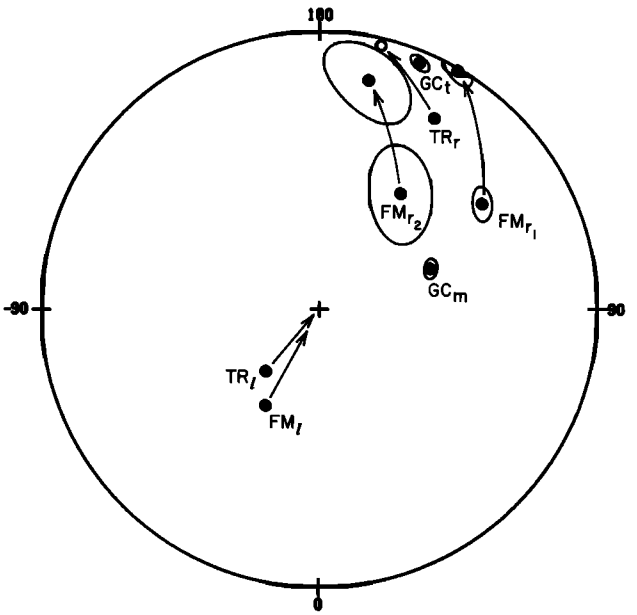


Fig. 9. Equal-area plot of paleomagnetic poles and confidence ovals, centered on north geographic poles. Solid circles on northern (upper) hemisphere; open circles on southern. Abbreviations of poles in Table 1. If the poles from the limestones (TR_ℓ and FM_ℓ) are made to coincide with the Tertiary (present-day) pole at the center of the plot (arrows), the poles from the red-brown unit (FM_{r1}, FM_{r2}, TR_r) change as shown by the curved arrows. These shifted poles correspond to the 'rotated poles' of Table 1.

in the part of the Muav studied by Elston and Bressler [1977] is clearly desirable. One possible test would be the relation of the hematite to calcite cement stratigraphy in these limestones. Cavities in limestones generally show a distinctive pattern of cement generations that reflect gradual infill of the cavity under different chemical conditions [e.g., Fuchtbauer, 1973]; this results in a stratigraphy of cements, with contrasting mophologies, that can allow relative dating of diagenetic events.

Red-Brown Unit, Bright Angel Shale

The low-inclination, hematite-borne magnetization in the red-brown unit of the Bright Angel Shale seems to be the best candidate for a Paleozoic magnetization. Petrographic study has shown that the hematite formed subsequently to deposition during oxidizing diagenesis. This oxidation did not go to completion, as much glauconite is preserved; this fact, together with the relative homogeneity of the magnetization, suggests that the oxidizing episode was relatively short. Very protracted oxidation would probably lead to scattered magnetizations like those seen in the Tapeats (discussed below). However, the time of oxidation spanned at least one polarity reversal. Finally, a return to reducing conditions followed, as indicated by the presence of authigenic ferroan rhombohedral carbonate, and this further supports a relatively short time for hematite formation.

A likely setting for the oxidizing diagenesis

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would occur if the red-brown unit acted as a perched aquifer, as it is a coarse-grained unit that is sandwiched between relatively impermeable, reduced shales. Cambrian strata in this area represent a classic marine transgressive sequence [McKee and Resser, 1945], and it seems unlikely that oxidizing diagenesis could penetrate to any significant depth in a subsiding continental margin. A number of major early Paleozoic marine regressions, the earliest probably of Late Cambrian age, are represented by a regional unconformity between Cambrian and Devonian strata [e.g., Palmer, 1971; Lochman-Balk, 1971]. Marine regressions would probably be the only favorable times to oxidize the red-brown unit, so the magnetization probably dates from the Late Cambrian at the earliest.

The magnetization directions from the red-brown unit are too scattered to determine if a significant declination difference exists between the Tramp Ridge and Frenchman Mountain sections.

Tapeats Sandstone and Sandy Interbeds, Lower Bright Angel Shale

The scattered magnetizations in the Tapeats and lower Bright Angel Shale reside in hematite. Similar scattered magnetizations, residing in hematite, have been attributed to multicomponent magnetizations that reside in hematite with high blocking temperatures, so that the components cannot be resolved by thermal demagnetization. Such components probably are CRM's (chemical remanent magnetizations) that result from protracted hematite authigenesis during polarity reversals [e.g., Roy and Park, 1974; Roy and Robertson, 1978; Van Alstine and Gillett, 1979a]. For both these units, such an explanation is consistent with the petrographic evidence of protracted diagenetic oxidation.

Elston and Bressler [1977] presented data on the Tapeats from sites in the eastern and central Grand Canyon and from central Arizona. The Tapeats in those sections is not white but brick red; whitish Tapeats like that at Frenchman Mountain, termed 'bleached' by those authors, is common in the Grand Canyon, but these authors focused their study on the less common red sections. Where sampled in the Grand Canyon, the Tapeats is of late Early and early Middle Cambrian age [McKee and Resser, 1945], and its age is probably similar in central Arizona.

In contrast to the results from Frenchman Mountain, this red Tapeats yields a consistent, two-polarity characteristic remanence that resides in hematite. The characteristic magnetizations were defined by bulk demagnetization of all samples at ~590°C after progressive demagnetization of pilot samples. In the Grand Canyon, both polarities are nearly antipodal and relatively well grouped. In central Arizona, the distributions are much more diffuse; Elston and Bressler [1977] ascribed the additional scatter in these areas to unresolved secondary components resulting from Devonian weathering, as there Devonian strata rest directly on the Tapeats. This hypothesis was not documented by the extraction of secondary components from demagnetization paths, however.

Elston and Bressler [1977] argued for penecontemporaneous origin of the two-polarity remanence based on its pole's differing from subsequent poles on the APW path, and on the presence of antipolar reversals. The first argument has been substantially weakened as a subsequently reported pole from Late Cambrian strata in Texas [Van der Voo et al., 1976; Watts et al., 1980b] lies within 2° of the Tapeats pole. Antipolar reversals are also a very weak criterion for primary remanence [e.g., Van Alstine and Gillett, 1979b]; antipolarity indicates that any remaining secondary components are small but yields no information about the time of acquisition of the two-polarity remanence.

Evidence suggesting that this two-polarity remanence was not acquired penecontemporaneously comes from consideration of the magnetization of the red-brown unit in southern Nevada. The red-brown unit is almost certainly coeval with the upper part of the Tapeats in the Grand Canyon where it was sampled by Elston and Bressler [1977], because the Tapeats is time transgressive, becoming younger west to east [McKee and Resser, 1945]. The directions of magnetization from the red-brown unit and the Tapeats differ (Table 1); however, the directions differ primarily in declination. If it is assumed that the magnetizations in the overlying limestones reflect a rotated Tertiary magnetization, the paleomagnetic poles for the red-brown unit that result after 'unrotating' the sections are not very different from the Grand Canyon Tapeats pole (Table 1 and Figure 9). This suggests that the direction of magnetization in the red-brown unit is the rotated equivalent of the Tapeats direction, a suggestion I had presented earlier in abstract [Gillett, 1979]. However, the petrographic study shows that the hematite pigment, and probably the characteristic magnetization, of the red-brown unit results from diagenetic oxidation. As a similar red pigment apparently carries the characteristic magnetization in the red Tapeats sampled by Elston and Bressler [1977], it seems possible that that magnetization also resides in authigenic hematite, perhaps dating from roughly the same time as the oxidation of the red-brown unit.

Conclusions

In Cambrian strata from cratonic sections in southern Nevada, in an area made structurally complicated by Mesozoic and Cenozoic tectonism, no primary component of magnetization was found despite investigation of a heterogeneous suite of rock units. Scattered magnetizations in the Tapeats Sandstone reside in hematite and appear to reflect vector sums of multiple, unresolved components of magnetization resulting from protracted diagenetic oxidation. In the red-brown unit of the overlying Bright Angel Shale, a consistent, two-polarity remanence also residing in hematite reflects an episode of oxidizing diagenesis, possibly during a Late Cambrian marine regression. This magnetization may be the rotated equivalent of that found in partly coeval, brick red Tapeats Sandstone in the central and eastern Grand Canyon. Finally, Middle Cambrian gray limestones carry a steep, normal polarity, magnetite-borne magnetization that may reflect either a late Mesozoic or Tertiary PTRM; they differ greatly in direction and magnetic

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carrier from similar, slightly younger limestones in the Grand Canyon.

This study, then, has little to contribute directly to the resolution of the problem of scattered Cambrian poles that was raised in the introduction. However, remagnetization of these strata was suggested by extensive magnetic work that was integrated with petrographic study and coupled with geochemical arguments. It seems possible that remagnetization could be an extremely common phenomenon in ancient rocks; further, structural ambiguities were demonstrated in this study that were not suspected from the geology. Before complex explanations for divergent paleomagnetic data are proposed, such as loops or hairpins in APW paths or nondipole behavior of the geomagnetic field, the more prosaic possibilities of remagnetization and structural rotation must be rigidly excluded.

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